# MD. Mehedi Hasan Maruf

Software Engineer

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## Summary

Computer Science engineer (RUET) with almost 4 years of industry experience building full-stack TypeScript/JavaScript systems, from Node.js/NestJS microservices to React/Next.js frontends. Currently engineering a TypeScript microservice backend at Aamira Software Solutions and building Omnious, a full-stack visual debugging platform on Fastify, tRPC, and Next.js. Also has hands-on DevOps experience with Docker, CI/CD, and Linux VPS management.

## Skills

**Programming Languages:** Python, JavaScript, TypeScript
**Web Technologies:** Node.js, NestJS, Express.js, React.js, Angular, Next.js, FastAPI, Django, .NET
**Databases:** MySQL, PostgreSQL, MongoDB, Supabase
**Tools & Concepts:** Git, Docker, CI/CD (GitHub Actions, Netlify), VPS (Linux) Management, REST APIs, ORM (Django ORM, Supabase), Pydantic, Project Management, Problem Solving
**AI/ML:** LangChain, Hugging Face Transformers, Semantic Analysis, NLP

## Experience

### Software Engineer II

***Jan 2025 – Present***

*Aamira Software Solutions*, Kokomo, Indiana, USA

- Led development of a customer relationship management system that increased client response rates by 30%.
- Architected and implemented a Typescript microservice backend solution that reduced API response time by 40%.
- Optimized database queries using Supabase, resulting in 60% improvement in application performance.

### Industrial Trainee

***Mar 2025***

*Brain Station 23*, Dhaka, Bangladesh

- Gained practical experience in enterprise software development methodology and practices.
- Served as Team Representative for a 15-person cohort, coordinating communication between management and trainees.
- Contributed to a student AI system using Next.js and Node.js that was deployed to an online server.

### Junior Software Engineer

***Nov 2022 – Apr 2024***

*Code Studio*, Rajshahi, Bangladesh

- Developed backends supporting thousands of daily users using JavaScript, TypeScript, and Python (Django).
- Managed 12 VPS (Linux) with 99.8% uptime and optimized CI/CD pipelines, reducing deployment time by 40%.
- Implemented Docker containerization, simplifying deployment across 5 different client environments.

## Personal Projects

### StudentAI - Intelligent Learning Companion

***Mar 2025***

*Read More:*

- Developed during industrial attachment at Brain Station 23: AI-powered learning assistant combining **FastAPI backend** with **Next.js frontend** for personalized education.
- Implemented a **RAG pipeline** with LangChain for document Q&A, semantic retrieval, and conversation memory; extended to YouTube content analysis and intelligent quiz generation.
- Built a **multimodal** document processing pipeline supporting PDFs, images, and videos with real-time WebSocket chat, JWT authentication, and progress tracking.
- Deployed production-ready application on Vercel (frontend) and Render (backend) with Supabase database integration.

### Omnious – Visual Debugging Platform *Nov 2025 – Present*

*Read More:*

- Building a full-stack visual debugging platform with interactive code-graph visualization, real-time trace animation, error heatmaps, and **AI-powered** chat-based debugging.
- Engineered a scalable **Node.js** backend using **Fastify 5** and **tRPC v11** with **TypeScript**, integrated with a **Next.js 16/React 19** frontend in a **Turborepo** monorepo.
- Integrated **PostgreSQL 15** with **pgvector**, **ltree**, and **pg_trgm** via **Supabase** for semantic search and hierarchical data; implemented Row-Level Security (RLS).
- Containerized the full-stack application using **Docker Compose** with multi-stage builds; automated deployment via **GitHub Actions** CI/CD pipeline.

## Education

### B.Sc. Engineering, Computer Science and Engineering *Mar 2022 – Aug 2026*

*Rajshahi University of Engineering & Technology (RUET)*, Rajshahi, Bangladesh

- CGPA: 3.75

## Certifications & Achievements

- Digital Accessibility Training (DAT) - a2i (Aspire to Innovate), Govt. of Bangladesh (Oct 2025)
- Industrial Training Certificate - Brain Station 23 (Mar 2025)
- PHP (Laravel) - EDGE Project (Feb 2025)
- Experience Certificate - Code Studio (Apr 2024)
- 4th Place - RUET CodeSmash (Team Competition) - Competitive Programming Contest (Mar 2023)
- 4th Place - RUET CodeSpark (Individual) - Competitive Programming Contest (Oct 2022)

## References

**Dr. Md. Shahid Uz Zaman**
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Profile

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**Faria Afrin**
Software Engineer, Samsung
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